EPISCOPE

MINISTRY POLICY BRIEF

District-Level Disease Surveillance in Bangladesh

Project: EpiScope – Disease Surveillance and Resource Allocation System
Course: CSE 4345 – Big Data Analytics

1. Introduction

Disease surveillance is important for identifying outbreaks and planning healthcare resources. In Bangladesh, health data may come from different sources and may contain missing values, duplicate records, and reporting delays. These problems can make disease monitoring difficult.EpiScope is an academic prototype developed to demonstrate how big data technologies can help organize health information, identify unusual disease patterns, forecast future cases, and support resource planning.

2. Main Features of EpiScope

The project covers 64 districts of Bangladesh and includes the following features:

- **Data Integration:** Combines district boundaries, population, weather, facility, and health-record data.
- **Data Quality:** Uses 17 quality rules to identify incorrect and incomplete records.
- **Outbreak Detection:** Checks weekly disease cases to identify unusual increases.
- **Disease Forecasting:** Forecasts dengue and diarrhoea cases for the next one to four weeks.
- **Resource Allocation:** Demonstrates how available resources can be distributed among healthcare facilities.
- **Dashboard:** Displays district-level information using five coordinated visual views.

3. Project Results

Measurement	Result
Districts	64
Synthetic facilities	640
Initial encounter records	10,200
Data quality rules	17
Quarantined records	349
Clean records	9,851

Large-scale test (SF=1)	25 million base rows
Total with duplicates	25.5 million

The outbreak detection system generated 64 alert records. It detected 7 out of 31 synthetic outbreak events, giving an event-level sensitivity of 22.58%. This shows that the current detection method needs improvement.

For forecasting, the project tested dengue and diarrhoea using the Seasonal Naive method. The forecasting results were useful for demonstrating the process, but further testing with other methods is needed.The resource allocation model distributed 494 synthetic demand units without unmet demand. The optimization model changed allocations at 30 facility-disease entries compared with the proportional method.

4. Recommendations for the Ministry

The following steps are recommended before using a system like EpiScope in real healthcare services:

1. **Improve Data Quality:** Use common formats and correct district and facility information.
2. **Improve Outbreak Detection:** Test different detection methods with real historical health data.
3. **Improve Forecasting:** Compare different forecasting models to obtain more reliable results.
4. **Better Resource Planning:** Use verified information about available beds, medicines, and diagnostic supplies.
5. **Protect Patient Information:** Use secure storage, access control, and privacy protection.
6. **Human Supervision:** Healthcare experts should review alerts and resource recommendations before taking action.

5. Implementation Plan

EpiScope can be developed further through four steps:

- **Step 1:** Prepare data standards and security policies.
- **Step 2:** Test the system using authorized historical health data.
- **Step 3:** Run a pilot alongside the existing surveillance process.
- **Step 4:** Expand the system only after proper testing and approval.

6. Limitations and Conclusion

EpiScope is an academic prototype. Its encounter records, facilities, outbreak events, and resource information are synthetic. The current outbreak detection and forecasting methods have not been validated for real public-health decisions.The project demonstrates how big data technologies can support disease monitoring and resource planning in Bangladesh. However, real-world implementation requires reliable health data, further testing, privacy protection, and expert supervision.EpiScope is a technical prototype and should not yet be treated as a fully operational public-health system.

Data Sources: DGHS district boundary data, population data attributed to BBS 2022, and Open-Meteo historical weather data. Synthetic datasets were generated for project testing.